

AHMED BARGOUGUI

Backend Engineer Intern | Software Engineer Intern | Data Engineering

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PROFILE

Backend Engineer Intern and Computer Science student building production-grade backend systems, authentication infrastructures, and data pipelines. Delivered scalable solutions used by **32,000+ concurrent users** and completed a data engineering internship at Sagemcom. Experienced in **REST API design, authentication systems, ETL pipelines, and scalable backend architecture**. Strong interest in distributed systems, system reliability, and backend performance optimisation. Seeking a **Backend or Software Engineering Internship — Summer 2026**.

PROFESSIONAL EXPERIENCE

Data Engineering Intern — Sagemcom • Tunis, Tunisia

07/2025 – 08/2025

- Designed and implemented a **Star Schema** data warehouse (5+ dimension tables), improving query performance for cross-team reporting
- Engineered **ETL pipelines** in Python (Pandas, SQLAlchemy) automating data ingestion from multiple business sources, reducing manual processing time
- Delivered **Power BI dashboards** enabling real-time KPI tracking for sales and operations — directly used by management
- Cleaned and standardised 10,000+ records using Power Query, improving data consistency and reporting accuracy
- Configured end-to-end data engineering environment: SQL Server, SSMS, Jupyter, Power BI — enabling full analytics workflow

Backend Engineer — Freelance — Online Gaming Platform • 32,000+ Active Users

02/2026 – 03/2026

- Architected and deployed a secure **REST API** (Node.js, Express, MySQL) supporting **32,000+** concurrent users in a production environment
- Designed scalable and high-performance backend architecture handling concurrent authentication requests with zero downtime
- Engineered device-based authentication (**HWID binding**) and OAuth2 integration — significantly reducing unauthorised access attempts
- Implemented **JWT session management** with token expiry, rotation, and refresh logic ensuring session integrity at scale
- Built a **C# desktop launcher** integrated with backend services to manage secure user access at application startup
- Applied rate limiting, API validation, and caching mechanisms to ensure system reliability and prevent abuse
- Managed full project lifecycle using Git version control — sole backend engineer responsible for architecture, implementation, and deployment

PROJECTS

ScanVD — AI-Powered Video Search Engine — Deployed on HuggingFace Spaces

2026

- Built a full-stack AI system extracting **precise timestamps** from natural language queries over video content
- Designed **multi-model backend pipeline** combining Whisper (ASR), YOLOv8 (object detection), and CLIP (semantic search)
- Deployed scalable inference API on HuggingFace Spaces — accessible at huggingface.co/spaces/AhmedBrg/ScanVD_demo

Stack: Python • Whisper • YOLOv8 • CLIP • HuggingFace • REST API • Docker

AgriLink — Collaborative Farming Platform — Hackathon Winner 🏆 • github.com/Ahmedbrg/Hackathon2025

2025

- Designed and shipped a mobile platform for AI-based plant disease detection — winner of the TGN **Hackathon 2025**
- Built GPS-based farmer networking with offline-first architecture for low-connectivity rural environments

Stack: React Native • TypeScript • AI API • GPS • Offline Architecture

Problem Solving — ACM/Codeforces — github.com/Ahmedbrg/problemsolving-HACKATHON

2026

- Ranked 3rd** in an ACM-style competitive programming contest — solved all problems under time pressure
- Practised algorithmic problem solving on Codeforces — arrays, graphs, dynamic programming, time complexity optimisation

Stack: Python • Algorithms • Data Structures • Competitive Programming

- Built a full-featured task manager with priorities, deadlines, categories, real-time search, and dark/light mode
 - Implemented bilingual UI (French/English) and deployed as a standalone web application
- Stack: *HTML · CSS · JavaScript · LocalStorage API*

TECHNICAL SKILLS

Backend:	Node.js · Express · REST APIs · Authentication Systems · JWT · OAuth2 · MySQL · C# · API Scalability
Data:	SQL Server · ETL · Pandas · SQLAlchemy · Power BI · Power Query · Star Schema · Data Modelling
AI / ML:	Python · Whisper · YOLOv8 · CLIP · HuggingFace · Computer Vision · Speech Recognition
Languages:	Python · JavaScript · Java · C++ · C# · SQL
Tools:	Git · GitHub · Linux · Docker · Postman · VS Code · Jupyter
Concepts:	System Design · Scalable Architecture · Concurrency · Database Optimisation · API Design · Version Control

EDUCATION

Bachelor's Degree in Computer Science — FSTSBZ — University of Kairouan, Tunisia09/2024 – 06/2027

Relevant Coursework: Algorithms · Data Structures · Databases · OOP · Operating Systems · Networks · Software Engineering

LEADERSHIP & ACTIVITIES

Founder & President — Club DevUp · FSTSBZ University09/2025 – Present

- Founded and lead a university tech club — organised workshops on AI/ML and LinkedIn career development
- Built a developer community promoting collaborative engineering culture and industry readiness among students

Trainer & Former Junior Secretary General — JCI Sidi Bouzid — Junior Chamber International01/2023 – Present

- Delivered training sessions on leadership, communication, and personal development to club members
- Progressed from Junior Secretary General to active Trainer — recognised for facilitation skills

LANGUAGES

Arabic NativeFrench ProfessionalEnglish Professional